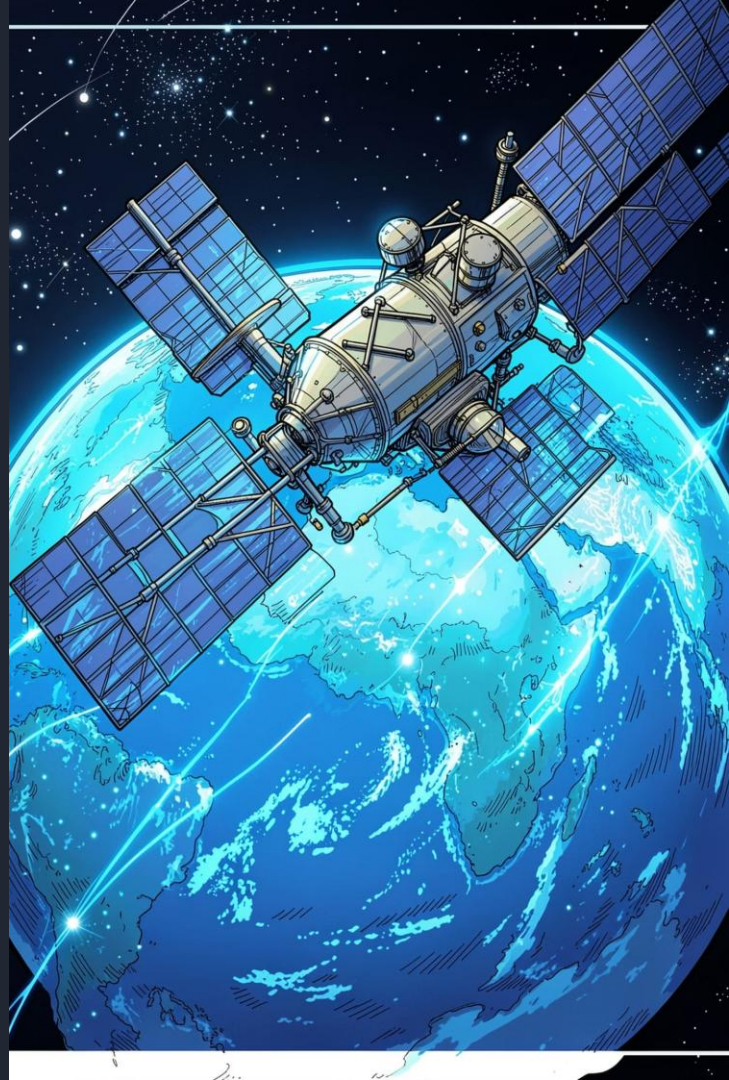


Smart Vision Satellite

An AI-powered satellite image analysis system running on **Raspberry Pi 5** — detecting objects, comparing images, and delivering fast, accurate insights for smarter decisions.

AI · SATELLITE IMAGING · RASPBERRY PI



The Problem We're Solving



Current Challenges

Manual analysis is **slow and error-prone**

Large areas are **difficult to monitor** consistently

Changes are detected **too late** to act

Decision-making is **delayed** without automated support

 **Our Solution:** Smart Vision Satellite automates image analysis and delivers results in seconds.

Project Objectives



Detect Objects

Identify buildings, roads, vehicles, forests, and more from satellite imagery.



Fast Decisions

Deliver accurate, actionable information to reduce response time.



Compare Images

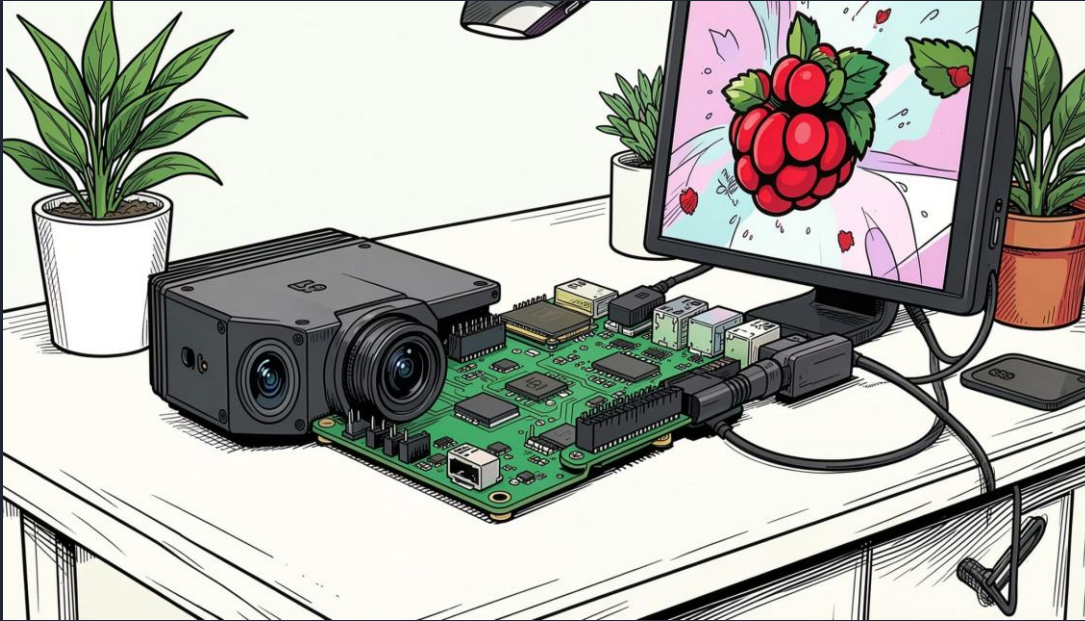
Analyze before-and-after satellite images to spot critical changes.



Save Effort

Automate detection to reduce human workload and eliminate manual errors.

Hardware Components



Component	Purpose
Raspberry Pi 5	Main processing unit
Web Camera	Capture or upload images
Display Screen	Show AI results
SD Card	Store operating system
Wi-Fi Module	Internet connectivity
Power Supply	Power the Raspberry Pi

The Raspberry Pi 5 is the brain of the system – compact, affordable, and powerful enough to run AI models at the edge.

Software & Technologies

Python

Core programming language for AI logic

OpenCV

Image processing and computer vision

PyTorch

Deep learning framework for model training

TorchGeo

Satellite and geospatial AI models

FastAPI

Lightweight backend API server

React

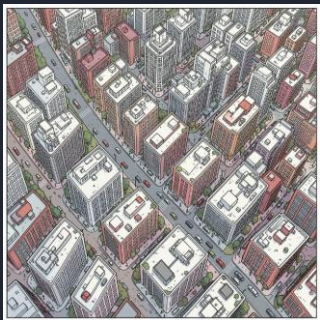
Responsive frontend user interface

How It Works



The pipeline is fully automated — from image input to AI-powered analysis and a clear, actionable report displayed in real time on the screen.

Objects the AI Can Detect



Urban Infrastructure

Buildings, roads, residential areas, and industrial zones



Natural Features

Forests, rivers, lakes, and agricultural farmland



Transportation

Vehicles, airports, ships, and road networks

Real-World Applications



Built for Impact

Smart Vision Satellite serves a wide range of critical sectors — from emergency responders tracking natural disasters to city planners designing tomorrow's infrastructure.

- Forest & environmental monitoring
- Smart city & urban development
- Disaster response & management
- Precision agriculture

Key Advantages



Fast Processing

AI delivers results in seconds



High Accuracy

Trained on diverse satellite datasets



Fully Automatic

No manual intervention required



Low Cost

Raspberry Pi keeps hardware affordable



Portable

Compact and easy to deploy anywhere



Smart Decisions

Actionable insights for stakeholders

What's Next & Conclusion

Future Roadmap

01

Live Satellite Connection — real-time data feeds

02

Drone Integration — aerial field deployment

03

Cloud Storage & Mobile App — remote access & alerts

04

AI Chat Assistant — conversational query interface

05

3D Satellite Mapping — immersive terrain visualization

In Summary

Smart Vision Satellite is a powerful, affordable AI system that **detects objects, compares images, and identifies changes** — empowering governments, NGOs, and researchers to make faster, smarter decisions. The future of satellite intelligence is **automated, accessible, and actionable**.